SOLUTION

Solution → Solution is a homologous mixture of two or more substances.

Binary solution
↓
2 component
↙ ↘
Solute — Solvent (generally present in larger quantity)

* Solution can be solid. eg: Alloys
  - brass (Cu + Zn)
  - Bronze (Cu + Sn)

* Solution can be gas as well. eg: Air.

Solute ← solid / liquid / gas Solvent ← solid / liquid / gas

- depending upon physical state of solute & solvent, solution can be of 9 types.

Table 1.1: Types of Solutions

Type of Solution	Solute	Solvent	Common Examples
Gaseous Solutions	Gas	Gas	Mixture of oxygen and nitrogen gases
	Liquid	Gas	Chloroform mixed with nitrogen gas
	Solid	Gas	Camphor in nitrogen gas
Liquid Solutions	Gas	Liquid	Oxygen dissolved in water
	Liquid	Liquid	Ethanol dissolved in water
	Solid	Liquid	Glucose dissolved in water
Solid Solutions	Gas	Solid	Solution of hydrogen in palladium
	Liquid	Solid	Amalgam of mercury with sodium
	Solid	Solid	Copper dissolved in gold

* physical state of solution is same as that of solvent.

Question

solute

Which of the following represents a liquid in gas solution?

└ solvent

(A) Oxygen dissolved in water

Chloroform + N_2

(B) Chloroform mixed with nitrogen gas

(C) Copper dissolved in gold

(D) Amalgam of mercury with sodium

Concentration terms –

1) $w/w\ \% = \frac{W_{solute}}{W_{solution}} \times 100$

2.) $w/v\ \% = \frac{W_{solute}}{V_{solution}} \times 100$

3.) $v/v\ \% = \frac{V_{solute}}{V_{solution}} \times 100$

4.) parts per million (ppm) = used when solute in traces

$1 million = 10 lakh = 10^6$

$ppm = \frac{W_{solute}}{W_{solution}} \times 10^6$

5.) Mole Fraction (X) $\quad \chi_B = \frac{n_B}{n_B + n_A} \qquad \chi_A = \frac{n_A}{n_A + n_B}$

A + B
solvent solute

$\chi_A + \chi_B = 1$

Question – Calculate the mole fraction of urea (NH_2CONH_2) in an aqueous solution containing 15% urea by mass. (w/w%)

solute – urea
solvent – water

Ans – 15 gm urea → 100 gm solution

$$\chi_{urea} = \frac{n_{urea}}{n_{urea} + n_{H_2O}}$$

$$n_{urea} = \frac{15}{60} = \frac{1}{4}$$

$$n_{H_2O} = \frac{85}{18}$$

$100 = 15 + w_{H_2O}$

$w_{H_2O} = 85$ gm.

$$\chi_{urea} = \frac{\frac{1}{4}}{\frac{1}{4} + 4.72} = \frac{0.25}{0.25 + 4.72} = \frac{0.25}{4.97}$$

Question

Which of the following statements is true for any given solution?

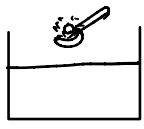
A. The sum of all mole fractions is greater than 1.

B. The sum of all mole fractions is less than 1.

C. The sum of all mole fractions is equal to unity. (1)

D. The sum of all mole fractions depends on the solute concentration.

Solubility of solids in liquids



- NaCl gets dissolved in water
- Naphthalene & anthracene do not dissolve in water

$$\text{Solute} + \text{solvent} \underset{\text{crystallization}}{\overset{\text{dissolution}}{\rightleftharpoons}} \text{Solution} \quad \Rightarrow \text{at equilibrium}$$

* Saturated solution – max^{m} amount of solid is dissolved.

° like dissolves like
- polar substance dissolves in polar solvent
- non polar substance dissolves in non polar solvent.

* Solubility of solid in liquid is equal to amount of solid dissolved in given amount of solvent at equilibrium.

Temperature affects solubility of solid in liquids

$\text{Solid} + H_2O(l) \rightarrow \text{solution}(aq.)$, ΔH = +ve (endothermic)

$T \uparrow \Rightarrow$ solubility $\uparrow$ Endo

$\text{Solid} + H_2O(l) \rightarrow \text{solution}(aq.)$, ΔH = −ve (exothermic)

$T \uparrow \Rightarrow$ solubility $\downarrow$ Exo

Pressure doesn't affect much solubility of solid in liquid.

Question

Read the passage and answer the following question:

The solubility of a substance depends on the nature of the substances involved. In addition to this, two other factors, temperature and pressure, also influence solubility. The solubility of a solid in a liquid is significantly affected by changes in temperature. According to Le Chatelier's Principle, if the dissolution process is endothermic ($\Delta_{sol} H > 0$), solubility increases with an increase in temperature, while if the process is exothermic ($\Delta_{sol} H < 0$), solubility decreases with an increase in temperature. Pressure, however, does not significantly affect the solubility of solids in liquids, as solids and liquids are highly incompressible and largely unaffected by pressure changes.

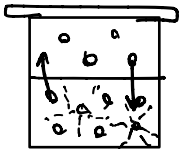
A What are the factors affecting the solubility of a substance?

B According to Le Chatelier's Principle, how does temperature affect the solubility of a solid in a liquid when the dissolution process is endothermic?

C What happens to the solubility of a solid in a liquid when the dissolution process is exothermic and temperature increases?

D Why doores pressure have a negligible effect on the solubility of solids in liquids?

Solubility of gases in liquids



at equilibrium, $Rate_u = Rate_d$

* solubility of gas is equal to amount of gas dissolved in given amount of liquid solvent at equilibrium at given temperature.

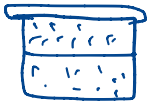
* On increasing temp, solubility of gas decreases in liquids.

  – fishes are uncomfortable in warm water as lesser O_2 is dissolved.

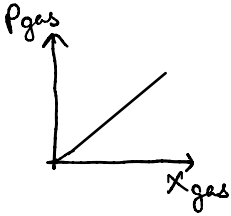
Effect of pressure on solubility of gas in liquid

Henry's law – It states that solubility of a gas in

Liquid is directly proportional to pressure of gas above liquid at equilibrium.



$P_{gas} \propto x_{gas}$



- K_H depends on nature of gas
- $K_H \propto \frac{1}{\text{solubility}}$
- K_H has units of pressure

  1 atm = 1 bar = 10^5 Pa
- He is least soluble gas.

$P_{gas} = K_H \times x_{gas} + 0$

$y = m \quad x \quad + c$

x_{gas} = mole fraction of gas in liquid.

K_H = Henry's constant.

Question

State Henry's law and explain its relationship between pressure and the solubility of a gas in a liquid.

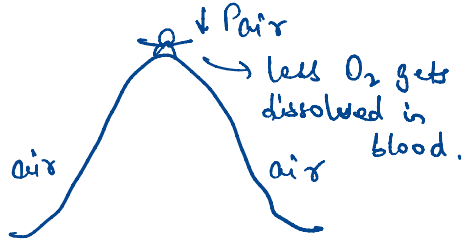
Ans: It states that pressure of gas over liquid is directly proportional to mole fraction of gas in liquid at equilibrium.

$P_{gas} \propto x_{gas}$

$P_{gas} = K_H \cdot x_{gas}$

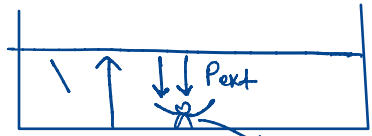
Applications of Henry's law

1) Anoxia: $P_{gas} \propto x_{gas}$



2.) Scuba diving – $He + N_2 + O_2$

(least soluble)



excess N_2 dissolved.

bents

Question

Low concentration of oxygen in the blood and tissues of people living at high altitude is due to :

(a) high atmospheric pressure

(b) low temperature

(c) low atmospheric pressure

(d) both low temperature and high atmospheric pressure

Question

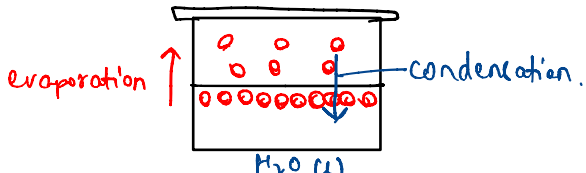
Question

Why do climbers and people living at high altitudes experience low oxygen levels in their blood and tissues?

At high altitude, due to low atmospheric pressure, lesser O_2 is dissolved in blood as per Henry's law

$P_{gas} \propto x_{gas}$

Vapour Pressure – It is the pressure exerted by vapours of a liquid over its liquid surface at equilibrium at a given temperature.

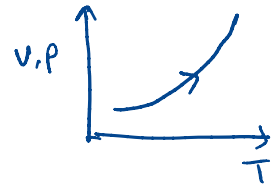


at equilibrium,

rate of evaporation = rate of condensation

vapour pressure of a liquid never changes except temperature.

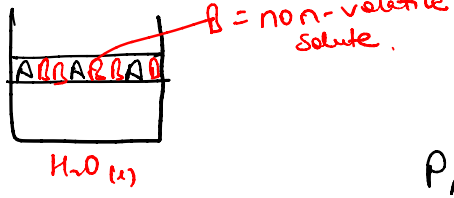
- On ↑ temp. V.P of a liquid ↑es.



- $H_2O_{(l)}$ < C_6H_6 $_{(l)}$

- more the intermolecular forces of attraction in liquid, lesser the V.P of liquid.

Effects of addition of non volatile solute to liquid.



$P_A < P^0_A$

P^0_A = V.P of pure solvent

P_A = V.P of solution / V.P of solvent in soln.

RAOULT'S LAW

$$P_A = P^0_A \cdot X_A$$

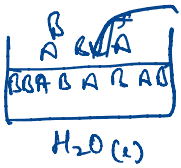
P_A = V.P of solvent in soln

P^0_A = V.P of pure solvent

X_A = mole fraction of solvent in solution.

Raoult's law states that V.P of a component in soln is given by product of V.P of pure component & mole fraction of component in solution.

volatile solute.



P°_A = V.P of pure solvent.

$P_{Total} = P_A + P_B$

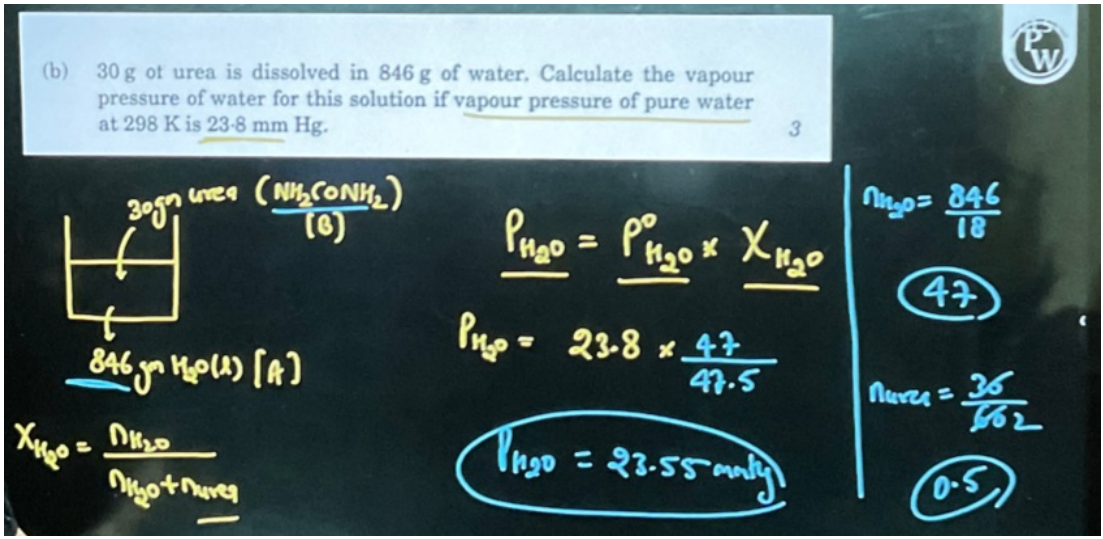
$P_A = P^\circ_A \cdot X_A$

$P_B = P^\circ_B \cdot X_B$

$P_{Total} = P^\circ_A X_A + P^\circ_B X_B$

$P^\circ_B > P^\circ_A$ $\quad P_{Total} > P^\circ_A$

$P^\circ_B < P^\circ_A$ $\quad P_{Total} < P^\circ_A$



On the basis of Roult's law, soln are of 2 types.

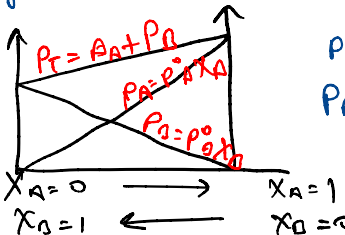
Ideal solutions

→ follow roult's law

A-A = A-B = B-B $\quad \Delta H_{mix} = 0$, $\Delta V_{mix} = 0$

force of attraction

eg: Benzene + toluene



$P_A = P^\circ_A X_A + 0$

$P_B = P^\circ_B X_B + 0$

Non-ideal solutions

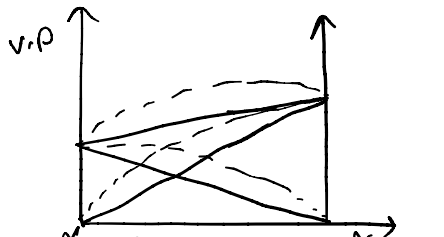
→ don't follow roult's law.

A-A ≠ A-B ≠ B-B

eg: $C_2H_5OH + H_2O$, $HNO_3 + H_2O$

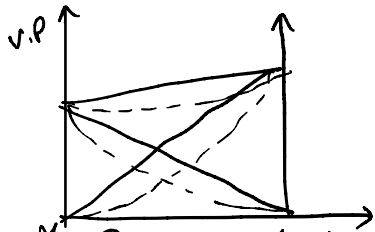
+ve deviation	−ve deviation
A-A > A-B	A-A < A-B
ΔH_{mix} = +ve	ΔH_{mix} = −ve
ΔV_{mix} = +ve	ΔV_{mix} = −ve

+ve deviation



A−A > A+B

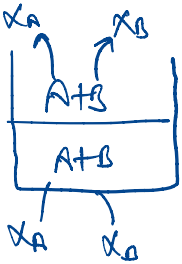
−ve deviation



A−A < A−B

Azeotropes

Those mixtures in which two liquids can't be separated by boiling / distillation.



- bcoz mole fraction in liquid phase & vapour phase remains same
- Non-ideal soln forms azeotrope

azeotropes

- max^m boiling azeotropes → +ve deviation
- min^m boiling azeotropes → −ve deviation

Question

Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): The enthalpy of mixing for ideal solutions is zero. ✓

Reason (R): In ideal solutions, the interactions between solute-solute, solvent-solvent, and solute-solvent are not equal. ✗

In the light of the above statements, choose the correct answer from the options given below:

(A) A is true but R is false. ✓

(B) A is false but R is true.

(C) Both A and R are true and R is the correct explanation of A.

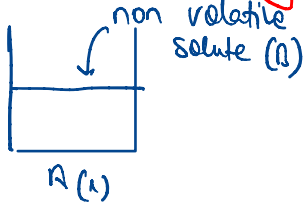
(D) Both A and R are true but R is NOT the correct explanation of A.

Colligative Properties — those properties of soln where value depends only upon no. of solute particles in solution.

4 colligative properties —

1) Relative lowering of V.P
2) Elevation in boiling point
3) Depression in freezing point
4) Osmotic pressure

1) Relative lowering of V.P.



$P^\circ_A \rightarrow P_A$

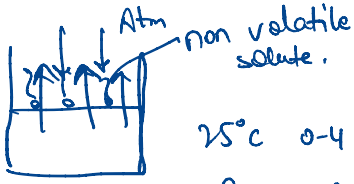
$$P_A = P^\circ_A X_A$$

$$\frac{P_A}{P^\circ_A} = X_A$$

$$1 - \frac{P_A}{P^\circ_A} = 1 - X_A$$

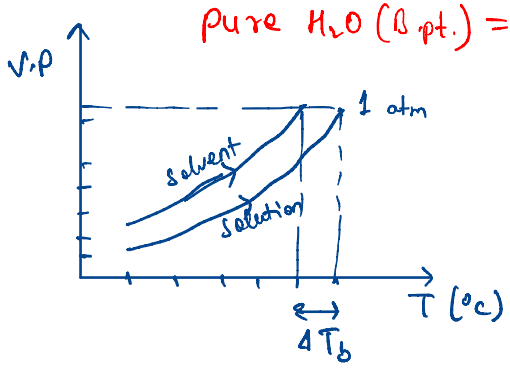
$$\frac{P^\circ_A - P_A}{P^\circ_A} = X_B$$

2.) Elevation in Boiling point. — addn of non-volatile solute increases the boiling point of soln as compared to Bpt. of pure solvent.



B.pt of a liquid is that temp. at which V.P of a liquid becomes equal to atmospheric pressure.

Pure H_2O (B.pt.) = 100°c



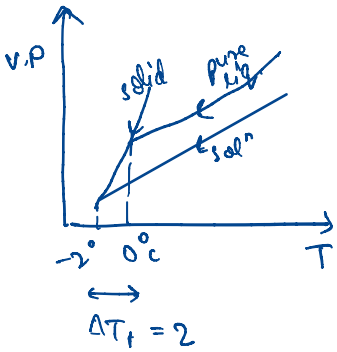
$\Delta T_b \propto m$

$\Delta T_b = K_b m$

* K_b depends on solvent

$$K_b = \frac{MRT_b^2}{1000\,\Delta H_{vap.}}$$

3.) Depression in freezing point. — freezing point of soln decreases as compared to f.pt of solvent when non volatile solute is added



$\Delta T_f \propto m$

$\Delta T_f = K_f m$ K_f = solvent

$$K_f = \frac{MRT_f^2}{1000\,\Delta H_{fusion.}}$$

$(H_2O)_{f.pt} = 0°c$

$f.pt = -5°c$

$\Delta T_f = 5$

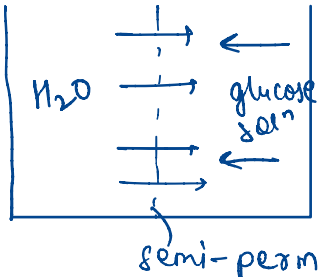
4.) Osmotic pressure — It is the pressure being exerted on a solution due to inward flow of solvent through SPM during osmosis

— It is the pressure required to stop osmosis.

$P = CRT$ $C = \frac{n}{V}$

$R = 0.0821$ atm/mol

— osmotic pressure can be measured at any temp.

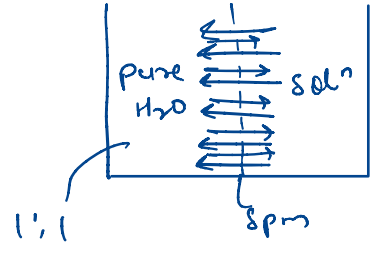


- Osmotic pressure is used to determine molecular masses of macromolecules.

Applications -

- mango → conc^n. NaCl sol^n

- Osmosis
  lower conc^n sol^n $\xrightarrow{\text{solvent}}$ higher conc^n cell.
- edema - lot of salt khate h.
- Reverse osmosis R.O



reverse flow of solvent by providing energy
= desalination.

Question -

The colligative property used for the determination of molar mass of polymers and proteins is :

(a) Osmotic pressure
(b) Depression in freezing point
(c) Relative lowering in vapour pressure
(d) Elevation is boiling point

Question –

Question

What happens to a raw mango when placed in a concentrated salt solution?

Vant Hoff factor (i) – if solute is electrolyte.

$1\ NaCl \rightarrow 1\ Na^+ + 1\ Cl^-$

$i = \dfrac{\text{no. of solute particles after dissociation}}{\text{no. of solute particles initially taken}}$

$i_{NaCl} = \dfrac{2}{1}$

$i_{NaCl} = 2$

$Na_2SO_4 \rightarrow 2Na^+ + SO_4^{2-}$

$i = \dfrac{3}{1}$ $\quad i = 3$

$i = \dfrac{\text{observed colligative property}}{\text{calculated value of colligative property.}}$

$\dfrac{P^\circ_A - P_A}{P^\circ_A} = i x_B$

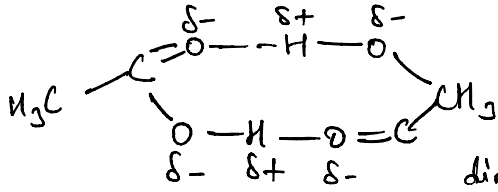
$\Delta T_b = i k_b m$

$\Delta T_f = i k_f m$

$P = iCRT$

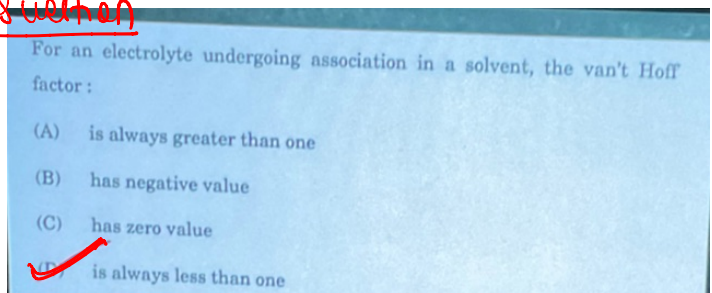
for dissociation $i > 1$

for association $i < 1$ $\quad i = \dfrac{1}{2}$



dimer.

Question



Question

Assertion (A) : When NaCl is added to water a depression in freezing point is observed. ✓

Reason (R) : NaCl undergoes dissociation in water. ✓

A & R correct but R is not correct explanation.

Question

Question

What is azeotrope? Difference between positive and negative azeotrope?

THANK YOU.